# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $\frac{1}{2}$ | favourable outcomes: 2, 4, 6 ⇒ 3 out of 6 |
| 2 | **(d)** $\frac{5}{6}$ | favourable outcomes: 1, 2, 3, 4, 6 ⇒ 5 out of 6 |
| 3 | **(c)** $\frac{2}{5}$ | 8 favourable out of 20 equally likely balls |
| 4 | **(a)** $\frac{3}{10}$ | 6 favourable out of 20 equally likely balls |
| 5 | **(b)** $\frac{2}{11}$ | 2 favourable out of 11 letters |
| 6 | **(b)** 47 | substitute $n = 10$ |
| 7 | **(c)** $n = 122$ | $5n - 3 = 607 \Rightarrow n = 122$ |
| 8 | **(b)** $n = 37$ | $7n + 2 = 261 \Rightarrow n = 37$ |
| 9 | **(d)** 98 | $a + (n-1)d = 2 + 24 \times (4) = 98$ |
| 10 | **(b)** 210 | $\frac{20 \cdot 21}{2}$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $1 - 0.35 = 0.65$, since the win and not-win events together are certain (probability 1). | 1+1 |
| 12 | (i) $\frac{4}{52} = \frac{1}{13}$. (ii) $\frac{2}{52} = \frac{1}{26}$. | 1+1 |
| 13 | $a = 3, d = 5$: $t_{10} = 3 + 9 \times 5 = 48$; $t_{26} = 3 + 25 \times 5 = 128$. | 1+1 |
| 14 | $21 - 3(n-1) = -81 \Rightarrow n = 35$ (1). $21 - 3(n-1) = 0 \Rightarrow n = 8$, so 0 is the 8th term (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | (i) $\frac{1}{3} \times 12 = 4$ (1). (ii) $\frac{6}{14} = \frac{3}{7}$ (2). | 1+2 |
| 16 | $2 \cdot 3^{n-1} = 486 \Rightarrow 3^{n-1} = 243 = 3^5 \Rightarrow n = 6$. | 2+1 |
| 17 | Subjective: a person's own opinion from what they see (‘sunny, so unlikely’), differing between people (1). Objective: based on evidence such as past rainfall data (relative frequency) or on equally likely outcomes (theoretical) (2). | 1+2 |
| | **OR:** (i) 2, 3, 5, 7 ⇒ $\frac{4}{8} = \frac{1}{2}$ (ii) 6, 7, 8 ⇒ $\frac{3}{8}$. | 1+1+1 |
| 18 | $18,\ 13.5,\ 10.125$ ft (2). It is a GP with $r = \frac{3}{4}$ (1). | 2+1 |
| | **OR:** Area at Stage $n$ is $\left(\frac{3}{4}\right)^n$ ⇒ Stage 3: $\frac{27}{64}$ (2). 27 triangles each of area $\frac{1}{64}$ give $\frac{27}{64}$ ✓ (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) HHH, HHT, HTH, THH, HTT, THT, TTH, TTT (2). (ii) 3 outcomes ⇒ $\frac{3}{8}$ (1). (iii) $1 - \frac{1}{8} = \frac{7}{8}$ (1). (iv) HHH, TTT ⇒ $\frac{2}{8} = \frac{1}{4}$ (1). | 2+1+1+1 |
| | **OR:** (i) $\frac{8}{200} = 0.04$ (1). (ii) $0.04 \times 5000 = 200$ (1). (iii) 0.96 (1). (iv) A bigger, representative sample varies less from the true proportion (2). | 1+1+1+2 |
| 20 | (a) AP, $d = 4$, $t_n = 4n + 3$ (1). (b) GP, $r = 3$, $t_n = 2 \cdot 3^{n-1}$ (1). (c) Neither (differences 3,5,7; ratios 4, 2.25) but $t_n = n^2$ (1). (d) GP, $r = \frac{1}{2}$, $t_n = 100\left(\frac{1}{2}\right)^{n-1}$ (2). | 1+1+1+2 |
| | **OR:** (i) $-5, -2, 1, 4, 7$ (1). (ii) $t_n = 3n - 8$ (1). (iii) $3n - 8 = 52 \Rightarrow n = 20$, yes (1). (iv) $\frac{20}{2}(-5 + 52) = 470$ (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET B

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 0.23 | $46/200 = 0.23$ |
| 2 | **(a)** 4 | multiply the number of outcomes of each experiment |
| 3 | **(d)** 600 | expected number $= P\times$ population |
| 4 | **(c)** 1 | always happens |
| 5 | **(b)** theoretical probability | Law of Large Numbers |
| 6 | **(d)** 96 | $ar^{n-1} = 3 \times (2)^5 = 96$ |
| 7 | **(c)** 64 | $ar^{n-1} = 1 \times (-2)^6 = 64$ |
| 8 | **(b)** 13 | $T_4 = 7$, $T_5 = 4 + 2 + 7 = 13$ |
| 9 | **(a)** neither an AP nor a GP | differences 2,3,4,5 are not constant |
| 10 | **(d)** $3^n$ | each black triangle becomes three |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | (i) $\frac{3}{20}$. (ii) $\frac{17}{20}$. | 1+1 |
| 12 | 36 equally likely outcomes; sum 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) ⇒ $\frac{6}{36} = \frac{1}{6}$. | 1+1 |
| 13 | $t_n = 11 + (n-1)(-3) = 14 - 3n$ (1). Recursive: $t_1 = 11,\ t_n = t_{n-1} - 3$ (1). | 1+1 |
| 14 | $S_{20} = \frac{20\times21}{2} = 210$; $S_{100} = \frac{100\times101}{2} = 5050$. | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | (i) 5, 10, 15, 20, 25 ⇒ $\frac{5}{25} = \frac{1}{5}$ (ii) 2,3,5,7,11,13,17,19,23 ⇒ $\frac{9}{25}$. | 1+2 |
| 16 | GP with $r = 2$: $500 \times 2^6 = 32000$. | 2+1 |
| 17 | $\frac{18}{30} = 0.6$ (1+1). Expected $= 0.6 \times 50 = 30$ (1). | 2+1 |
| | **OR:** For few trials the relative frequency can differ much from the theoretical value, but as the number of trials grows it tends to get closer to the theoretical probability (e.g. tosses of a fair coin approach 0.5). | 3 |
| 18 | $1, 2, 3, 5, 8, 13, 21, 34, 55, 89$. | 3 |
| | **OR:** $1, 5, 13, 29, 61, 125, 253, \ldots$ (2). 133 does not appear and the terms increase, so it is not a term (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) A personal judgement, e.g. 'it is sunny so I think rain is unlikely'; different people may differ (2). (ii) Count the days with rain in past years and compute the relative frequency (2). (iii) Rain depends on many sensitive atmospheric factors, so the outcome cannot be predicted with certainty (1). | 2+2+1 |
| | **OR:** (i) Primes 2,3,5,7 ⇒ $\frac{4}{8} = \frac{1}{2}$ (1). (ii) Composites 4,6,8 ⇒ $\frac{3}{8}$ (1). (iii) Not fair, $\frac{1}{2} \neq \frac{3}{8}$ (1). (iv) 1 (neither prime nor composite), probability $\frac{1}{8}$ (2). | 1+1+1+2 |
| 20 | (i) Points $(1,1), (2,5), (3,9), (4,13), (5,17)$ lie on the line $y = 4x - 3$ (2). (ii) $(1,3), (2,6), (3,12), (4,24), (5,48)$ do not lie on a line; slopes 3, 6, 12, 24 keep rising (2). (iii) An AP grows by a fixed amount (linear), a GP by a fixed multiple (much faster) (1). | 2+2+1 |
| | **OR:** (i) 100, 150, 200 (1). (ii) $t_n = 100 + 50(n-1) = 50n + 50$ (1). (iii) $t_{12} = 650$; total $= \frac{12}{2}(100 + 650) = ₹4,500$ (2). (iv) $50n + 50 = 1000 \Rightarrow n = 19$ (1). | 1+1+2+1 |

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## ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $\frac{1}{3}$ | favourable outcomes: 3, 6 ⇒ 2 out of 6 |
| 2 | **(a)** $\frac{1}{3}$ | favourable outcomes: 3, 5 ⇒ 2 out of 6 |
| 3 | **(b)** $\frac{3}{5}$ | 9 favourable out of 15 equally likely balls |
| 4 | **(d)** $\frac{3}{5}$ | 9 favourable out of 15 equally likely balls |
| 5 | **(a)** $\frac{2}{11}$ | 2 favourable out of 11 letters |
| 6 | **(c)** 38 | substitute $n = 7$ |
| 7 | **(b)** $n = 69$ | $2n - 1 = 137 \Rightarrow n = 69$ |
| 8 | **(a)** 48 | $a + (n-1)d = 3 + 9 \times (5) = 48$ |
| 9 | **(d)** 39 | $a + (n-1)d = -5 + 11 \times (4) = 39$ |
| 10 | **(d)** 1275 | $\frac{50 \cdot 51}{2}$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Multiples of 3: 3, 6 ⇒ $\frac{2}{8} = \frac{1}{4}$. | 1+1 |
| 12 | Both outcomes are equally likely ($\frac{1}{2}$ each) and no one can predict the result beforehand, so neither team has an advantage. | 2 |
| 13 | Ratios $\frac{6}{2} = \frac{18}{6} = \frac{54}{18} = 3$, a constant (1). $t_n = 2 \cdot 3^{n-1}$ (1). | 1+1 |
| 14 | ₹240, ₹280, ₹320 — an AP with $d = 40$ (1). $t_n = 200 + 40n$; $t_{10} =$ ₹600 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | (i) $\frac{2}{52} = \frac{1}{26}$ (ii) $\frac{1}{52}$ (iii) $\frac{36}{52} = \frac{9}{13}$. | 1+1+1 |
| 16 | $3,\ 6,\ 30,\ 870$. | 3 |
| 17 | (i) $\frac{24}{40} = \frac{3}{5}$ (ii) $\frac{16}{40} = \frac{2}{5}$ (and $\frac{3}{5} + \frac{2}{5} = 1$). | 1+1+1 |
| | **OR:** Only 2 is both even and prime ⇒ $\frac{1}{6}$ (1). The numbers that are even or prime are 2, 3, 4, 5, 6, so only 1 is neither ⇒ $\frac{1}{6}$ (2). | 1+2 |
| 18 | Differences $2, 3, 4$ — not constant ⇒ not an AP (1+). Ratios $3, 2, \frac{5}{3}$ — not constant ⇒ not a GP (1). | 1+1+1 |
| | **OR:** $t_n = 3n - 1$ is linear: points $(1, 2), (2, 5), (3, 8)$ lie on $y = 3x - 1$ (2). Slope $= 3$, the common difference (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $\frac{5}{1000} = 0.005$ (1). (ii) $\frac{10 \times 5}{1000} = 0.05$ (2). (iii) $0.995$ (1). (iv) $0.005 \times 200 = 1$ (1). | 1+2+1+1 |
| | **OR:** (i) win $0.6$, draw $0.15$, lose $0.25$ (2). (ii) $0.6 + 0.15 + 0.25 = 1$ ✓ (1). (iii) $0.6 \times 30 = 18$ (1). (iv) No. It is only an estimate from past data; probability tells what happens in the long run, not what will happen next (1). | 2+1+1+1 |
| 20 | (i) 1000, 2000, 4000, 8000 (1). (ii) $500 \cdot 2^n$ (1). (iii) $500 \times 64 = 32000$ (1). (iv) $500 \cdot 2^n > 10000 \Rightarrow 2^n > 20 \Rightarrow n = 5$ (16,000) (2). | 1+1+1+2 |
| | **OR:** (i) AP with $a = 1$, last term $2n - 1$: $S = \frac{n}{2}(1 + 2n - 1) = n^2$ (2). (ii) 25 terms ⇒ $25^2 = 625$ (1). (iii) $1 + 3 + \cdots + 23 = 12^2 = 144$ (2). | 2+1+2 |

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## ANSWER KEY & MARKING SCHEME SET D

Chapters 7 & 8: **The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 0.15 | $6/40 = 0.15$ |
| 2 | **(d)** 12 | multiply the number of outcomes of each experiment |
| 3 | **(a)** 120 | expected number $= P\times$ population |
| 4 | **(c)** 0.65 | $1 - 0.35$ |
| 5 | **(d)** Jñān-Chaupaḍ | as given in the chapter |
| 6 | **(b)** 162 | $ar^{n-1} = 2 \times (3)^4 = 162$ |
| 7 | **(d)** 29 | 1, 5, 13, 29 |
| 8 | **(d)** an AP with $d = 3$ | constant difference 3 |
| 9 | **(a)** 55 | $1 + 2 + \cdots + 10$ |
| 10 | **(c)** Prakrit metre and poetry | 7th century CE, Vṛttajātisamuchaya |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\frac{112}{200} = 0.56$. The theoretical value is $0.5$; with more tosses the experimental value tends to get closer. | 1+1 |
| 12 | 7 months (Jan, Mar, May, Jul, Aug, Oct, Dec) out of 12: $\frac{7}{12}$. | 1+1 |
| 13 | Yes — every term is $-1$ times the previous one: $r = -1$, $t_n = (-1)^{n-1}$. | 1+1 |
| 14 | Stage $n$ has $3^n$ black triangles: Stage 4 has 81 and Stage 5 has 243. | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Weekend (Sat, Sun) ⇒ $\frac{2}{7}$ (1). Days beginning with T: Tuesday, Thursday ⇒ $\frac{2}{7}$ — equally likely (2). | 1+2 |
| 16 | Slope between first two $= 3$; between the next two $= 6$ (2). Unequal slopes ⇒ not collinear (1). | 2+1 |
| 17 | $\frac{520}{1000} = 0.52$ (1+1). Difference from $0.5$ is $0.02$ (1) — small because the number of trials is large. | 2+1 |
| | **OR:** April, June, September, November ⇒ $\frac{4}{12} = \frac{1}{3}$ (3). | 3 |
| 18 | $S = 1 + 2 + \cdots + n$ and $S = n + (n-1) + \cdots + 1$ (1). Adding: $2S = n(n+1)$ since there are $n$ pairs, each $n+1$ (1). So $S = \frac{n(n+1)}{2}$ (1). | 1+1+1 |
| | **OR:** $t_n + t_{n-1} = \frac{n(n+1)}{2} + \frac{(n-1)n}{2} = n^2$ (2). $55 + 45 = 100 = 10^2$ ✓ (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $\frac{13}{52} = \frac{1}{4}$ (1). (ii) $\frac{39}{52} = \frac{3}{4} = 1 - \frac{1}{4}$ (2). (iii) $\frac{26}{52} = \frac{1}{2}$ (1). (iv) 8 cards ⇒ $\frac{8}{52} = \frac{2}{13}$ (1). | 1+2+1+1 |
| | **OR:** (i) $\frac{10}{50} = 0.2$ (1). (ii) bus $0.36$, bicycle $0.2$, walking $0.24$, car $0.2$; sum $= 1$ (2). (iii) $0.24 \times 1200 = 288$ (1). (iv) Bus is most likely (0.36); least likely: bicycle/car (0.2), difference $0.16$ (1). | 1+2+1+1 |
| 20 | (i) $S_{58} - S_{24} = 29 \times 59 - 12 \times 25 = 1711 - 300 = 1411$ (2). (ii) 34 terms; $\frac{25+58}{2} \times 34 = 1411$ ✓ (2). (iii) Āryabhaṭa, in the Āryabhaṭīya, Chapter 2, verse 19 (1). | 2+2+1 |
| | **OR:** (i) $\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n(2n)}{2} = n^2$ (2). (ii) $55 + 45 = 100$ ✓ (1). (iii) $t_{17} = 153$, $t_{80} = 3240$ (2). | 2+1+2 |

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